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Inventor(s)

GUADALUPI; Carlo et al.

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### MEASURING SYSTEM FOR MEASURING THE LEVEL OF TONER IN A CARTRIDGE FOR LASER PRINTER

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#### Abstract

A device includes an acceleration sensor, a charge variation sensor and processing circuitry. The acceleration sensor, in operation, generates an acceleration signal indicative of vibrations associated with a cartridge of a laser printer. The charge variation sensor, in operation, generates a charge-variation signal indicative of a variation of electrostatic charge associated with the cartridge of the laser printer. The processing circuitry is coupled to the acceleration sensor and to the charge variation sensor. The processing circuitry, in operation, generates feature parameters based on the acceleration signal and the charge-variation signal, and generates a level signal indicative of a level of toner in the cartridge based on the generated feature parameters. The generated feature parameters include a peak-to-peak distance of the acceleration signal and a maximum of the charge variation signal.

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**Inventors:** GUADALUPI; Carlo (Sedriano, IT), RIVOLTA; Stefano Paolo (Desio, IT), LABOMBARDA; Andrea (Milano, IT)

**Applicant:** STMicroelectronics International N.V. (Geneva, CH)

**Family ID:** 1000008496379

**Assignee:** STMicroelectronics International N.V. (Geneva, CH)

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## Background/Summary

### BACKGROUND

#### Technical Field

[0001] The present disclosure relates to a measuring system for measuring the level of toner in a cartridge for laser printer. Furthermore, it relates to a cartridge comprising the measuring system, to a laser printer comprising the cartridge, to a measuring method implemented by the measuring system and to a corresponding computer program product.

#### Description of the Related Art

[0002] Laser printers are a known and now widespread technology for printing on sheets of paper.

[0003] In particular, laser printing uses toner as a pigment, contained in a powder state in a cartridge present in the laser printer.

[0004] The image to be reproduced, coming from a paper or electronic support, is imprinted by a laser onto a selenium cylinder made photosensitive, referred to as “OPC drum,” which is electrically charged with the light radiation in such a way as to acquire the image in a negative manner and to transfer the toner onto the paper, where the latter is then fixed by fusion.

[0005] Accurately detecting the amount of toner present in the cartridge is therefore a critical action to ensure the correct functioning of the laser printer. In fact, this allows the user to always be informed about the actual level of toner available for printing, so as to allow the cartridge to be replaced in the most suitable manner (i.e., only when the toner is almost exhausted, in order to minimize waste and costs, but at the same time without the risk of not having the spare part available when necessary).

[0006] Several known solutions are currently available to measure the level of the toner in the cartridge. For example, it is known to use an inductive sensor for detecting the level of toner powder.

[0007] However, the known solutions do not ensure accurate and reliable measurements (in particular, several critical issues are known in their production which may cause issues in the final product) and usually do not allow the level of toner to be detected precisely (generally the measured level is indicated to the user as “full,” “medium” or “low”).

### BRIEF SUMMARY

[0008] In an embodiment, a device comprises: an acceleration sensor, which, in operation, generates an acceleration signal indicative of vibrations associated with a cartridge of a laser printer; a charge variation sensor, which, in operation, generates a charge-variation signal indicative of a variation of electrostatic charge associated with the cartridge of the laser printer; and processing circuitry coupled to the acceleration sensor and to the charge variation sensor; wherein the processing circuitry, in operation: generates feature parameters based on the acceleration signal and the charge-variation signal; and generates a level signal indicative of a level of toner in the cartridge based on the generated feature parameters. The generated feature parameters include a peak-to-peak distance of the acceleration signal and a maximum of the charge variation signal.

[0009] In an embodiment, a system comprises: a laser printer cartridge having an external body; and toner-level measurement circuitry coupled to the external body, wherein the toner-level measurement circuitry, in operation: generates an acceleration signal indicative of vibrations associated with the laser printer cartridge; generates a charge-variation signal indicative of a variation of electrostatic charge associated with the laser printer cartridge; generates feature

parameters based on the acceleration signal and the charge-variation signal; and generates a level signal indicative of a level of toner in the laser printer cartridge based on the generated feature parameters. The generated feature parameters include a peak-to-peak distance of the acceleration signal and a maximum of the charge variation signal.

[0010] In an embodiment, a method comprises: generating an acceleration signal indicative of vibrations associated with a laser printer cartridge; generating a charge-variation signal indicative of a variation of electrostatic charge associated with the laser printer cartridge; generating feature parameters based on the acceleration signal and the charge-variation signal; and generating a level signal indicative of a level of toner in the laser printer cartridge based on the generated feature parameters. The generated feature parameters include a peak-to-peak distance of the acceleration signal and a maximum of the charge variation signal.

[0011] In an embodiment, a non-transitory computer-readable medium's contents configure processing circuitry to perform a method, the method comprising: generating an acceleration signal indicative of vibrations associated with a laser printer cartridge; generating a charge-variation signal indicative of a variation of electrostatic charge associated with the laser printer cartridge; generating feature parameters based on the acceleration signal and the charge-variation signal; and generating a level signal indicative of a level of toner in the laser printer cartridge based on the generated feature parameters. The generated feature parameters include a peak-to-peak distance of the acceleration signal and a maximum of the charge variation signal.

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## Description

### BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0012] For a better understanding of the present disclosure, embodiments are now described, purely by way of non-limiting example, with reference to the attached drawings, wherein:

[0013] FIG. 1 is a schematic cross-sectional view of a cartridge for a laser printer, according to one embodiment;

[0014] FIG. 2 is a perspective schematic view, in ghost, of a laser printer comprising the cartridge of FIG. 1, according to one embodiment;

[0015] FIG. 3 is a block diagram schematically showing details of a measuring system for measuring the level of toner of the cartridge of FIG. 1, according to one embodiment;

[0016] FIG. 4 is a block diagram that schematically shows classification criteria for measuring the level of toner of the cartridge of FIG. 1, according to one embodiment;

[0017] FIG. 5 shows an example of a confusion matrix obtained through the classification criteria of FIG. 4; and

[0018] FIG. 6 is a block diagram that schematically shows a measuring method performed by the measuring system of FIG. 3, according to one embodiment.

[0019] In the following description, elements common to the different embodiments have been indicated with the same reference numbers.

### DETAILED DESCRIPTION

[0020] FIG. 1 shows a cartridge 10 for a laser printer, shown in FIG. 2 with the reference 40.

[0021] The cartridge 10 is configured to contain the toner 12, in particular powder of toner 12, which in use is used to print on the sheet of paper.

[0022] The general structure of the cartridge 10 is of a known type.

[0023] In particular, the cartridge 10 comprises an external body 14 (e.g., a container) which accommodates elements such as a toner hopper 16, an agitator bar 18, a developer roller 20, an OPC drum 22, a primary charge roller (PCR) 24, a wiper blade 26 and a waste bin 28.

[0024] The toner hopper 16 contains the toner 12 to be used for printing. In detail, the powder of toner 12 is kept in movement inside the toner hopper 16 through the agitator bar 18, also present

inside the toner hopper **16** and configured to rotate in such a way as to continuously move the toner **12**.

[0025] The developer roller **20** is positioned in an opening of the toner hopper **16** and is interposed between the toner hopper **16** and the OPC drum **22**. Rotating axially in a controlled manner, the developer roller **20** allows the selective transfer of the toner **12** onto the OPC drum **22**.

[0026] The PCR **24** is in contact with the OPC drum **22** and is configured to negatively charge an external photosensitive surface of the OPC drum **22**, whereon the toner **12** is then transferred through the developer roller **20**.

[0027] Through a laser opening **30** present in the external body **14**, a laser beam **32** may enter the external body **14** and impinge on the OPC drum **22**, in such a way as to imprint, on the external photosensitive surface of the OPC drum **22**, an image which is the negative of the image to be printed. In particular, this prevents the toner **12** from adhering to the external photosensitive surface of the OPC drum **22** at the hidden image.

[0028] Furthermore, the wiper blade **26** is in contact with the OPC drum **22** in such a way as to remove any residues of the toner **12** which may have remained erroneously stuck to the OPC drum **22** after the contact of the latter with the sheet to be printed, without therefore having been transferred to the sheet. These residues of the toner **12** detached through the wiper blade **26** end up in the waste bin **28**, having the wiper blade **26** faced thereto.

[0029] In use and in a per se known manner, laser printing comprises the following steps in succession to each other.

[0030] A static charging step of the OPC drum **22** is performed through the PCR **24**. In detail, a continuous flow of electrical current coming from the PCR **24** generates a negative charge state on the photosensitive surface of the OPC drum **22**.

[0031] In an exposure step, the laser beam **32** scans the surface of the OPC drum **22** removing the negative charge in the areas that will not have to be printed, i.e., whereon the toner **12** will not have to stick in the subsequent development step. Accordingly, at this point a hidden image, equal to the reproduction in a negative manner of the image to be printed on the sheet, is present on the surface of the OPC drum **22**.

[0032] In the development step the deposition of the powder of toner **12** is performed on the hidden image of the OPC drum **22**. The powder of toner **12**, coming from the toner hopper **16**, is electrically attracted to the OPC drum **22** and sticks to the areas remained sensitive to the negative charge.

[0033] In a transfer step, the OPC drum **22** slides on the sheet, which is electrically charged in a positive manner by a transfer roller of the laser printer **40**, to cause the toner **12** to be attracted by the sheet of paper and transferred thereon creating the image to be printed.

[0034] In a fusion step, the particles of toner **12** placed on the sheet of paper are fused thereon. In detail, the sheet of paper is passed through a part of the laser printer **40** referred to as “fusion unit,” comprising an upper fuser roller and a lower pressure roller. The lower pressure roller compresses the sheet onto the upper fuser roller, which is heated in such a way as to fuse the toner **12** sticking it to the paper.

[0035] In a subsequent cleaning step, the toner **12** remaining on the OPC drum **22** after the passage of the sheet of paper is cleaned away to allow the machine to repeat the printing process.

[0036] Finally, in an erasing step, the cartridge **10** erases the traces of negative charge still present on the developer roller **20**.

[0037] FIG. 2 shows the laser printer **40**, comprising the cartridge **10** and further elements that are known and therefore not described in detail, such as a drive roller **42** which carries the sheet to be printed (here indicated with the reference **41**) to the cartridge **10**, the transfer roller **44** which electrically charges in a positive manner the sheet **41** at where the latter contacts the OPC drum **22**, the fusion unit **46** with the upper fuser roller **46a** and the lower pressure roller **46b** for fusing the toner **12** to the sheet **41**, and a laser assembly **48** which comprises a laser emitter **48a** for emitting

the laser beam **32** and a rotating mirror **48b** to direct the laser beam **32** into the laser opening **30** of the cartridge **10**.

[0038] As previously described, the cartridge **10** has several rotating elements, which in use rotate for example in an axial manner around respective rotation axes (e.g., the OPC drum **22**, the developer roller **20**, the PCR **24**, etc.). Accordingly, the cartridge **10** is configured to generate vibrations during its use, which are measurable.

[0039] Furthermore, due to its rotation, the agitator bar **18** generates, in use, a movement of the particles of toner **12** in order to prevent them from compacting with each other and then making it difficult for them to transfer towards the OPC drum **22**. However, since the toner **12** contains about 5-6% of ferromagnetic material (e.g., iron oxide), it has been verified that the movement of the toner **12** through the agitator bar **18** generates an electrostatic charge variation which is also measurable.

[0040] In particular, and as better described below, the Applicant has verified that the vibrations and the electrostatic charge variations generated by the cartridge **10** are correlated to the amount of toner **12** present in the toner hopper **16**. Accordingly, the level of toner **12** in the cartridge **10** may be measured by analyzing the measurements of vibration and/or electrostatic charge variation generated by the cartridge **10**.

[0041] For this purpose, and as better shown in FIG. **3**, the cartridge **10** comprises a measuring system **50** for measuring the level of the toner **12** in the cartridge **10**.

[0042] The measuring system comprises an acceleration sensor **52**, a charge variation sensor **54**, or both an acceleration sensor **52** and a charge variation sensor **54**.

[0043] The acceleration sensor **52** comprises an accelerometer, for example a triaxial accelerometer, configured to measure accelerations corresponding to the vibrations generated, in use, by the rotating elements of the cartridge **10**.

[0044] In particular, the acceleration sensor **52** generates, in use, an acceleration signal indicative of the vibrations generated by the cartridge **10**. For example, the acceleration signal comprises, at each instant, a respective acceleration value for each acceleration component measured by the acceleration sensor **52**, e.g., a respective acceleration value for each of the measuring axes of the accelerometer.

[0045] For illustrative and non-limiting purposes, the acceleration sensor **52** may comprise an accelerometer LIS2DUXS12 of the present Applicant, for example with an “output data rate” (ODR) equal to about 200 Hz.

[0046] In detail, the acceleration sensor **52** is carried by the external body **14** of the cartridge **10**, it is fixed thereto. For example, the acceleration sensor **52** is arranged externally to the external body **14**, it is fixed to an external surface of the external body **14**.

[0047] In greater detail, the acceleration sensor **52** is fixed to a region of the external body **14** of the cartridge **10** which is placed at the rotating elements of the cartridge **10** (e.g., at the OPC drum **22**). For example, the region of the external body **14** having the acceleration sensor **52** fixed thereto can be the part of the external body **14** that is closest to the rotating elements of the cartridge **10**. This is exemplarily shown in FIG. **1**.

[0048] The charge variation sensor **54** is configured to measure the charge variations, in detail electrostatic charge, generated by the movement of the toner **12** due to the agitator bar **18**.

[0049] In particular, the charge variation sensor **54** generates, in use, a charge variation signal indicative of the electrostatic charge variations generated by the cartridge **10**.

[0050] For illustrative and non-limiting purposes, the charge variation sensor **54** may comprise a LIS2DUXS12 of the present Applicant, for example with an “output data rate” (ODR) equal to about 200 Hz.

[0051] In detail, the charge variation sensor **54** is carried by the external body **14** of the cartridge **10**, it is fixed thereto. For example, the charge variation sensor **54** is arranged externally to the external body **14**, it is fixed to the external surface of the external body **14**.

[0052] In greater detail, the charge variation sensor **54** is fixed to a region of the external body **14** of the cartridge **10** which is placed at the toner hopper **16** and, more in detail, at the agitator bar **18**. For example, the region of the external body **14** having the charge variation sensor **54** fixed thereto can be the part of the external body **14** which is closest to the agitator bar **18** of the cartridge **10**. This is exemplarily shown in FIG. 1.

[0053] In detail, the charge variation sensor **54** comprises one or more electrodes (not shown). Hereinafter, reference is exemplarily made to the case of an electrode.

[0054] The electrode of the charge variation sensor **54** is placed in the cartridge **10** as previously described. In detail, the electrode is arranged facing the toner hopper **16** and may be electrically shielded towards the outside of the cartridge **10** (e.g., a metal part of the electrode extends between the external body **14** and an insulating layer that electrically insulates the electrode from the environment surrounding the cartridge **10**, making it sensitive exclusively, or in any case mainly, to electrostatic charge variations generated by the cartridge **10**).

[0055] In use, the electrode detects the electrostatic charge variations and generates the corresponding charge variation signal indicative of the measured electrostatic charge variations.

[0056] Furthermore, the measuring system **50** comprises a control unit or circuit **56** configured to receive the acceleration signal from the acceleration sensor **52** and/or the charge variation signal from the charge variation sensor **54** and, on the basis of the acceleration signal and/or the charge variation signal, generate a level signal indicative of the level of the toner **12** in the cartridge **10** (in detail, of the amount of toner **12** present in the toner hopper **16** and available for printing).

[0057] Hereinafter the case in which the control unit or circuit **56** is external to the acceleration sensor **52** and to the charge variation sensor **54** and is operationally coupled (in detail, electrically coupled) to the acceleration sensor **52** and to the charge variation sensor **54** is exemplarily considered. This case is shown in FIG. 3.

[0058] However, according to a different embodiment the control unit **56** may be a control unit integrated in the acceleration sensor **52** and/or in the charge variation sensor **54**. For example, the control unit **56** may be internal to the acceleration sensor **52** and may be operationally coupled to the charge variation sensor **54** in order to acquire the charge variation signal from the latter. This allows the potentialities of the electronic circuitry included in modern sensors to be fully exploited, without the need to use a control unit external to the sensors.

[0059] The control unit **56** is an electronic unit or circuit (such as a CPU, a microprocessor, a microcontroller or a dedicated calculation unit) which may comprise, coupled to each other, units of a known type and not shown such as: an interface unit or circuit to interface the acceleration sensor **52** and the charge variation sensor **54** with the control unit **56** (e.g., comprising an amplification stage and/or an analog-to-digital converter, ADC, stage); a processing unit or circuit for processing signals from the acceleration sensor **52** and the charge variation sensor **54**; and a data storage unit or memory (such as a memory, e.g., a non-volatile memory) for storing the acquired data.

[0060] In detail, the control unit **56** is carried by the external body **14** of the cartridge **10**, e.g., it is fixed thereto. For example, the control unit **56** is arranged externally to the external body **14**, it is fixed to the external surface of the external body **14**. This is exemplarily shown in FIG. 1. Furthermore, and in a manner not shown, the control unit **56** may be coupled to the acceleration sensor **52** and to the charge variation sensor **54** through electrical connections (e.g., electrical cables).

[0061] As shown in FIG. 3, the control unit **56** (e.g., the processing unit of the control unit **56**) comprises: a pre-processing module or circuit **60** (optional), a filtering module or circuit **62** (optional), a feature extraction module or circuit **64** and a classification module or circuit **66**.

[0062] The pre-processing module **60** is coupled to the acceleration sensor **52** to receive the acceleration signal from the latter. The pre-processing module **60** is configured to pre-process the acceleration signal, in detail calculating at each acquisition instant of the acceleration signal a

respective norm of the acceleration values considered on the measuring axes of the accelerometer in such a way as to have the (pre-processed) acceleration signal which is independent of the position and spatial orientation of the acceleration sensor **52**.

[0063] The filtering module (also referred to as charge variation filtering module) **62** is coupled to the charge variation sensor **54** to receive the charge variation signal from the latter. The filtering module **62** is configured to filter the charge variation signal, for example through a band-pass filter having a lower cut-off frequency comprised between about 0.1 Hz and about 1 Hz and for example equal to about 0.5 Hz and an upper cut-off frequency comprised between about 5 Hz and about 20 Hz and for example equal to about 5 Hz. In this manner the (filtered) charge variation signal does not depend on factors such as charge variations induced by the power supply at 50 Hz or 60 Hz present in closed environments such as the interiors of buildings or used to power the printer **40**.

[0064] The feature extraction module **64** is coupled to the pre-processing module **60** and/or to the filtering module **62** to receive from the latter the acceleration signal (pre-processed) and/or the charge variation signal (filtered), respectively. The feature extraction module **64** is configured to generate feature parameters indicative of features of the acceleration signal and/or the charge variation signal.

[0065] For illustrative and non-limiting purposes, the features considered are one or more of the following features: mean, variance, energy, “zero cross,” “peak detector,” peak-to-peak distance, minimum, maximum. For example, variance, “zero cross,” “peak detector” and peak-to-peak distance may be used; mean, “peak detector,” peak-to-peak distance, minimum, maximum may be used. For example, in the considered case both the acceleration signal and the charge variation signal are used the peak-to-peak distance may be used for the acceleration signal and the maximum may be used for the charge variation signal, as well as possibly also other features that have been previously mentioned for the two signals.

[0066] Accordingly, the feature extraction module **64** is configured to calculate the chosen features of the acceleration signal and/or the charge variation signal and to generate the feature parameters accordingly.

[0067] For example, the calculation of the features may be done on the basis of the acquisition of a few hundred samples of the acceleration signal and the charge variation signal (e.g., about 255), considering the ODR of the acceleration sensor **52** and the charge variation sensor **54** equal to about 200 Hz.

[0068] The classification module **66** is coupled to the feature extraction module **64** to receive from the latter the feature parameters of the acceleration signal and/or the charge variation signal. The classification module **66** is configured to generate the level signal on the basis of the feature parameters of the acceleration signal and/or the charge variation signal, therefore it is configured to determine the level of the toner **12** in the cartridge **10** as a function of the feature parameters.

[0069] In detail, the level signal may assume a value comprised between a plurality of defined or threshold values corresponding to a respective plurality of defined levels of toner **12** in the cartridge **10** (in detail, in the toner hopper **16**). For example, the value of the level signal may vary between five possible defined or threshold values, for example equal to 0%, 25%, 50%, 75%, and 100%. Each of these values identifies a respective level class to which the amount of toner **12** detected through the acceleration signal and/or the charge variation signal may belong. For example, if the acceleration signal and/or the charge variation signal are indicative of a percentage amount (e.g.,  $x$ ) of toner **12** in the toner hopper **16** comprised in a first variation range (e.g.,  $0\% \leq x < 12.5\%$ ), then the level signal assumes a first value (e.g., 0%) indicative of a first level class; if the percentage amount is comprised in a second variation range (e.g.,  $12.5\% \leq x < 37.5\%$ ), then the level signal assumes a second value (e.g., 25%) indicative of a second level class; if the percentage amount is comprised in a third variation range (e.g.,  $37.5\% \leq x < 62.5\%$ ), then the level signal assumes a third value (e.g., 50%) indicative of a third level class; if the percentage amount is comprised in a fourth variation range (e.g.,  $62.5\% \leq x < 87.5\%$ ), then the level signal assumes a

fourth value (e.g., 75%) indicative of a fourth level class; and if the percentage amount is comprised in a fifth variation range (e.g.,  $87.5\% \leq x \leq 100\%$ ), then the level signal assumes a fifth value (e.g., 100%) indicative of a fifth level class.

[0070] In detail, the classification module **66** generates, in use, the level signal on the basis of the feature parameters using machine learning or artificial intelligence techniques. In detail, the classification module **66** implements a classification algorithm, for example a “random forest” (RF) classifier, a “decision tree” (DT) classifier, or a “support vector machine” (SVM) classifier, etc. Hereinafter the case in which the classification module **66** implements a DT classifier is exemplarily considered.

[0071] The classifier is trained in a supervised manner and according to per se known techniques in order to recognize, given the features parameters received at input, which of the predefined level classes the measurement carried out belongs to, so as to choose the value of the level signal accordingly.

[0072] For example, in case only the acceleration signal is used, the DT classifier may have a number of leaves equal to about 6 and a tree size equal to about 11; in case only the charge variation signal is used, the DT classifier may have a number of leaves equal to about 11 and a tree size equal to about 21; and in case both the acceleration signal and the charge variation signal are used, the DT classifier may have a number of leaves equal to about 6 and a tree size equal to about 11.

[0073] In general, three different embodiments of the measuring system **50** previously described are described.

[0074] In a first embodiment, the measuring system **50** comprises the acceleration sensor **52** while the charge variation sensor **54** is absent. In this case, the control unit **56** does not comprise the filtering module **62**, therefore the feature extraction module **64** and the classification module **66** function only on the basis of the acceleration signal and the resulting feature parameters.

[0075] In a second embodiment, the measuring system **50** comprises the charge variation sensor **54** while the acceleration sensor **52** is absent. In this case, the control unit **56** does not comprise the pre-processing module **60**, therefore the feature extraction module **64** and the classification module **66** function only on the basis of the charge variation signal and the resulting feature parameters.

[0076] In a third embodiment, the measuring system **50** comprises both the acceleration sensor **52** and the charge variation sensor **54**. In this case, the control unit **56** may comprise both the filtering module **62** and the pre-processing module **60**, and in general the feature extraction module **64** and the classification module **66** function on the basis of both the acceleration signal and the charge variation signal (therefore in “data-fusion” mode), and the resulting feature parameters.

[0077] In particular, in the third embodiment, the feature extraction module **64** can calculate the peak-to-peak distance of the acceleration signal (e.g., after pre-processing by the pre-processing module **60**) and the maximum of the charge variation signal (e.g., the filtered charge variation signal outputted by the filtering module **62**). Moreover, the classification module **66** can use the DT classifier as a classification algorithm (e.g., with a number of leaves equal to about 6 and a tree size equal to about 11).

[0078] In this case, the DT classifier is based on classification criteria shown in FIG. 4.

[0079] In particular, in step **D1**, the DT classifier checks if the peak-to-peak distance of the acceleration signal (also called in the following distance A.sub.pp, for brevity) satisfies a first condition, for example, whether A.sub.pp is lower than, or equal to, a first threshold distance **A1**. For example, the first threshold distance **A1** can be equal to about 0.306 mg, to about 0.306396 mg.

[0080] If the first condition is satisfied (exit **Y** from **D1**, e.g., the distance A.sub.pp is lower than, or equal to, the first threshold distance **A1**), in step **D2** the DT classifier checks if the distance A.sub.pp satisfies a second condition, for example, whether A.sub.pp is lower than, or equal to, a second threshold distance **A2**. For example, the second threshold distance **A2** can be equal to about 0.141 mg, to about 0.141358 mg.



[0081] If the second condition is satisfied (exit Y from D2, e.g., the distance A.sub.pp is lower than, or equal to, the second threshold distance A2), in step D3 the DT classifier determines (assesses) that the level of the toner **12** in the cartridge **10** corresponds to a first toner level T1, for example, equal to about 100%.

[0082] If the second condition is not satisfied (exit N from D2, e.g., the distance A.sub.pp is greater than the second threshold distance A2), in step D4 the DT classifier checks if the maximum of the (filtered) charge variation signal (also called in the following maximum C.sub.m, for brevity) satisfies a third condition, for example, whether C.sub.m is lower than, or equal to, a first threshold maximum C1. For example, the first threshold maximum C1 can be equal to about 1.463 LSB, to about 1.463375 LSB.

[0083] If the third condition is not satisfied (exit N from D4, e.g., the maximum C.sub.m is greater than the first threshold maximum C1), in step D6 the DT classifier determines that the level of the toner **12** in the cartridge **10** corresponds to a third toner level T3, for example, equal to about 50%.

[0084] If the third condition is satisfied (exit Y from D4, e.g., the maximum C.sub.m is lower than, or equal to, the first threshold maximum C1), in step D5 the DT classifier checks if the distance A.sub.pp satisfies a fourth condition, for example whether A.sub.pp is lower than, or equal to, a third threshold distance A3. For example, the third threshold distance A3 can be equal to about 0.193 mg, to about 0.193359 mg.

[0085] If the fourth condition is satisfied (exit Y from D5, i.e. the distance A.sub.pp is lower than, or equal to, the third threshold distance A3), in step D7 the DT classifier determines that the level of the toner **12** in the cartridge **10** corresponds to the third toner level T3, in particular equal to about 50%.

[0086] If the fourth condition is not satisfied (exit N from D5, i.e. the distance A.sub.pp is greater than the third threshold distance A3), in step D8 the DT classifier determines that the level of the toner **12** in the cartridge **10** corresponds to a second toner level T2, in particular equal to about 75%.

[0087] If the first condition is not satisfied (exit N from D1, i.e. the distance A.sub.pp is greater than the first threshold distance A1), in step D9 the DT classifier checks if the distance A.sub.pp satisfies a fifth condition, in detail is lower than, or equal to, a fourth threshold distance A4. In particular, the fourth threshold distance A4 is equal to about 0.460 mg, in particular to about 0.46045 mg.

[0088] If the fifth condition is satisfied (exit Y from D9, i.e. the distance A.sub.pp is lower than, or equal to, the fourth threshold distance A4), in step D10 the DT classifier determines that the level of the toner **12** in the cartridge **10** corresponds to a fourth toner level T4, in particular equal to about 25%.

[0089] If the fifth condition is not satisfied (exit N from D9, i.e. the distance A.sub.pp is greater than the fourth threshold distance A4), in step D11 the DT classifier determines that the level of the toner **12** in the cartridge **10** corresponds to a fifth toner level T5, in particular equal to about 0%.

[0090] Therefore, the classification criteria of FIG. 4 allow to classify the level of the toner **12** in the cartridge **10** according to five predefined values, i.e. T1-T5.

[0091] In particular, the following relationship exists between the predefined toner values T1-T5:  $T1 > T2 > T3 > T4 > T5$ .

[0092] Moreover, the following relationship exists between the threshold distances A1-A4:  $A2 < A3 < A1 < A4$ .

[0093] FIG. 5 shows an example of a confusion matrix that can be obtained based on the third embodiment and using the classification criteria of FIG. 4. In particular, the indexes a-c correspond to the toner levels T5-T1.

[0094] As it can be deduced from FIG. 5, the use of the data from both the acceleration sensor **52** and the charge variation sensor **54** greatly and surprisingly increases the accuracy of the classification up to about 94.191%. In fact, it has been verified that the acceleration sensor **52** alone

has a low accuracy and sensitivity in the evaluation of the difference between the toner levels 50% and 75%, but the charge variation sensor **54** provides additional information that allow to perform this distinction with high accuracy and sensitivity. Thus, the combined use of these two sensors greatly increases the measurement reliability.

[0095] FIG. **6** shows a measuring method **80** performed by the control unit **56** to detect the level of the toner **12** in the cartridge **10**.

[0096] At a step **S10** of the measuring method **80**, the control unit **56** acquires the acceleration signal from the acceleration sensor **52**.

[0097] At a step **S12** of the measuring method **80**, the control unit **56** acquires the charge variation signal from the charge variation sensor **54**.

[0098] Similarly to what has been previously described with reference to the measuring system **50**, steps **S10** and **S12** may be performed individually or in combination with each other. Hereinafter the case in which the measuring method **80** uses both the acceleration signal and the charge variation signal to perform the detection of the level of toner **12** is exemplarily described.

[0099] At a step **S14** (optional) of the measuring method **80**, the control unit **56** pre-processes the acceleration signal, in detail through the pre-processing module **60** previously described and thus calculating the norm of the components of the acceleration signal.

[0100] At a step **S16** (optional) of the measuring method **80**, the control unit **56** filters the charge variation signal, in detail through the filtering module **62** previously described.

[0101] At a step **S18** of the measuring method **80**, the control unit **56** determines the feature parameters on the basis of the acceleration signal (in detail, pre-processed) and the charge variation signal (in detail, filtered), in detail through the feature extraction module **64** previously described.

[0102] At a step **S20** of the measuring method **80**, the control unit **56** generates the level signal on the basis of the feature parameters, in detail through the classification module **66** previously described.

[0103] The level signal may be provided to the user, for example through a visual or audible message generated by a user interface (not shown) coupled to the control unit **56** and controlled by the latter, or may be used to control by the control unit **56** one or more functionalities of the cartridge **10** or the printer **40** (or, possibly, of other electronic apparatuses coupled to the control unit **56**).

[0104] Furthermore, the measuring method **80** is performed by the control unit **56** through a suitable computer program product, stored in the control unit **56** and designed in such a way that, when executed in the control unit **56**, the latter becomes configured to perform the measuring method **80**.

[0105] From an examination of the characteristics of the disclosure made according to the present disclosure, the advantages that it affords are evident.

[0106] In detail, the measuring system **50** and the measuring method **80** allow the level of toner **12** present in the cartridge **10** of the laser printer **40** to be accurately and precisely detected, in such a way as to provide the user with a reliable indication of the amount of toner **12** still available for printing.

[0107] The measuring system **50** comprises elements with high measuring reliability and low cost (from both the financial and energetic point of view), therefore it allows the level of the toner **12** to be measured in a reliable, economical manner and with reduced energy consumption.

[0108] In particular, both of the two measuring channels (the acceleration sensor **52** and the charge variation sensor **54**) may be used in order to minimize the computational, energy and economical cost of the measuring system **50**, or both the measuring channels may be used in order to maximize measuring accuracy.

[0109] In fact, considering for exemplary purposes the number of leaves and branches previously indicated as the size of the classification tree of the DT classifier, it has been verified that the embodiment of the measuring system **50** which comprises the sole acceleration sensor **52** allows an

accuracy equal to about 84% to be achieved, while the embodiment of the measuring system **50** which comprises the sole charge variation sensor **54** allows an accuracy equal to about 86% to be achieved. On the other hand the embodiment of the measuring system **50** which comprise both the acceleration sensor **52** and the charge variation sensor **54** allows an accuracy equal to about 94% to be achieved (about 8-10% more than the accuracy obtained when only one of the sensors **52** and **54** is used. Furthermore, it has been verified that, when using both measuring channels, the computational complexity (e.g., the complexity of the DT classifier, for example in terms of features used and tree size) decreases with respect to the case of using the sole charge variation sensor **54** and is completely comparable to the case of using the sole acceleration sensor **52**, thus decreasing the overfitting of the classifier. This shows the correlation between the acceleration signal and the charge variation signal, due to the dependence of both on the level of toner **12** in the cartridge **10**.

[0110] Moreover, when using both the sensors **52** and **54**, such high values of accuracy can be achieved also by means of the two previously described features (A.sub.pp and C.sub.m); since a reduced number of features suffices for obtaining these high values of accuracy, the computational resources and time/power consumptions are also minimized.

[0111] Furthermore, the measuring system **50** is easily integrable into currently known cartridges **10** and does not require significant modifications to the latter.

[0112] Finally, it is clear that modifications and variations may be made to the disclosure described and illustrated herein without thereby departing from the scope of the present disclosure, as defined in the attached claims.

[0113] For example, the different embodiments described may be combined with each other to provide further solutions.

[0114] Furthermore, the pre-processing module **60** may be absent, for example because the calculation of the norm is already performed internally to the acceleration sensor **52** and therefore the acceleration signal at the output of the acceleration sensor **52** is already indicative of the norm of the measurements carried out on the measuring axes of the accelerometer. Furthermore, the same calculation of the norm may not be performed either by the pre-processing module **60** or by the acceleration sensor **52**, for example because the accelerometer present in the acceleration sensor **52** is of uniaxial type or is placed, with respect to the cartridge **10**, in such a way as to have only one axis which is stimulated by the vibrations induced by the cartridge **10**. In this case, the feature extraction module **64** may operate directly on the signal measured by the accelerometer, therefore not on the basis of the norm of the measurements.

[0115] Furthermore, the filtering module **62** may be absent. In this case, the filtering of the charge variation signal may be performed internally to the charge variation sensor **54** or it may not be performed. In this second case, the feature extraction module **64** may operate directly on the signal measured by the electrode, therefore not on the basis of the filtered signal.

[0116] Furthermore, although the feature extraction module **64** previously described is coupled to both the pre-processing module **60** and the filtering module **62** and therefore is configured to determine the feature parameters both on the basis of the acceleration signal and on the basis of the charge variation signal, it becomes however apparent that the feature extraction module **64** may be formed by a first feature extraction module (not shown) coupled only to the pre-processing module **60** and therefore configured to determine first feature parameters only on the basis of the acceleration signal, and by a second feature extraction module (not shown) coupled only to the filtering module **62** and therefore configured to determine second feature parameters only on the basis of the charge variation signal. In this case, the first and the second feature extraction modules are both coupled to the classification module **66**, which receives, in use, both the first and the second feature parameters in order to generate the level signal similarly to what has been previously described.

[0117] Furthermore, and in a manner not shown, the pre-processing module **60** may also implement

a filter configured to filter the acceleration signal in order to remove any noise or select particular frequency components of the acceleration signal. In detail, this filtering may be performed and, following the filtering, the acceleration signal (filtered) may be pre-processed as previously described. For example, this filtering is performed through a further band-pass filter having a lower cut-off frequency comprised between about 0.5 Hz and about 10 Hz and for example equal to about 5 Hz and an upper cut-off frequency comprised between about 50 Hz and about 100 Hz and for example equal to about 80 Hz.

[0118] Alternatively, this filtering of the acceleration signal may also be performed externally to the pre-processing module **60** and therefore through a further filtering module (also referred to as acceleration filtering module, not shown) which is coupled to the acceleration sensor **52** to receive from the latter the acceleration signal and is also coupled to the pre-processing module **60** to provide the latter with the acceleration signal which has been filtered and which may be pre-processed as previously described.

[0119] In an embodiment, a measuring system (**50**) for a cartridge (**10**) for laser printer (**40**), the measuring system (**50**) being configured to be coupled to an external body (**14**) of the cartridge (**10**) and including at least one of:—an acceleration sensor (**52**) configured to measure vibrations of the cartridge (**10**) generated by rotating elements (**20, 22, 24**) of the cartridge (**10**) and dependent on a level of toner (**12**) contained in the cartridge (**10**); and a charge variation sensor (**54**) configured to measure electrostatic charge variations generated by movements of the toner (**12**) in the cartridge (**10**) and dependent on the level of the toner (**12**) in the cartridge (**10**), the measuring system (**50**) further including a control unit (**56**) configured to: receive from the acceleration sensor (**52**) an acceleration signal indicative of the measured vibrations, and/or from the charge variation sensor (**54**) a charge variation signal indicative of the measured electrostatic charge variations; determine feature parameters as a function of the acceleration signal and/or the charge variation signal; and generate, on the basis of the feature parameters, a level signal indicative of the level of the toner (**12**) in the cartridge (**10**).

[0120] In an embodiment, the control unit (**56**) includes: a feature extraction module (**64**) coupled to the acceleration sensor (**52**) and/or to the charge variation sensor (**54**) and configured to determine the feature parameters; and a classification module (**66**) coupled to the feature extraction module (**64**) and configured to generate the level signal.

[0121] The control unit (**56**) can be integrated into one of the acceleration sensor (**52**) and the charge variation sensor (**54**) and can be operationally coupled to the other of the acceleration sensor (**52**) and the charge variation sensor (**54**), or the control unit (**56**) can be external to the acceleration sensor (**52**) and the charge variation sensor (**54**) and can be operationally coupled to the acceleration sensor (**52**) and to the charge variation sensor (**54**).

[0122] In an embodiment, a cartridge (**10**) for laser printer (**40**), includes an external body (**14**) and a measuring system (**50**), the measuring system (**50**) being coupled to the external body (**14**).

[0123] In an embodiment, rotating elements (**20, 22, 24**) of the cartridge can include one or more of: a developer roller (**20**), an OPC drum (**22**), and a primary charge roller (**24**), accommodated in the external body (**14**), wherein the cartridge (**10**) can further include a toner hopper (**16**) configured to contain the toner (**12**) and accommodated in the external body (**14**), and an agitator bar (**18**) extending into the toner hopper (**16**) and configured to rotate in the toner hopper (**16**) in such a way as to move the toner (**12**) into the toner hopper (**16**), wherein the acceleration sensor (**52**) can be at the rotating elements (**20, 22, 24**) and the charge variation sensor (**54**) can be at the toner hopper (**16**).

[0124] In an embodiment, a laser printer (**40**) includes a cartridge (**10**).

[0125] In an embodiment, a measuring method (**80**) for measuring a level of toner (**12**) in a cartridge (**10**) for laser printer (**40**), is performed through a measuring system (**50**) configured to be coupled to an external body (**14**) of the cartridge (**10**) and including at least one of: an acceleration sensor (**52**) configured to measure vibrations of the cartridge (**10**) generated by rotating elements

(20, 22, 24) of the cartridge (10) and dependent on the level of toner (12) in the cartridge (10); and a charge variation sensor (54) configured to measure electrostatic charge variations generated by movements of the toner (12) in the cartridge (10) and dependent on the level of the toner (12) in the cartridge (10), the measuring system (50) further including a control unit (56), the measuring method (80) including the steps of: receiving (S10, S12), by the control unit (56), from the acceleration sensor (52) an acceleration signal indicative of the measured vibrations, and/or from the charge variation sensor (54) a charge variation signal indicative of the measured electrostatic charge variations; determining (S18), by the control unit (56), feature parameters as a function of the acceleration signal and/or the charge variation signal; and generating (S20), by the control unit (56) and on the basis of the feature parameters, a level signal indicative of the level of the toner (12) in the cartridge (10).

[0126] The step of generating (S20) the level signal can include selecting a value of the level signal among a plurality of predefined values corresponding to a respective plurality of predefined levels of toner (12) in the cartridge (10).

[0127] The step of generating (S20) the level signal can include processing the feature parameters through a classifier, in particular of the decision-tree type.

[0128] The step of determining (S18) the feature parameters can include calculating, on the basis of the acceleration signal and/or the charge variation signal, one or more of the following features: mean, variance, energy, “zero cross,” “peak detector,” peak-to-peak distance, minimum, maximum.

[0129] The measuring method can further include the step of pre-processing (S14) the acceleration signal by calculating a norm of the acceleration signal, and the step of determining (S18) the feature parameters may include determining the feature parameters as a function of the acceleration signal that has been pre-processed.

[0130] The measuring method can further include the step of filtering (S16) the charge variation signal, and the step of determining (S18) the feature parameters can include determining the feature parameters as a function of the charge variation signal that has been filtered.

[0131] The step of filtering (S16) the charge variation signal can include using a band-pass filter having a lower cut-off frequency between 0.1 Hz and 1 Hz and an upper cut-off frequency between 5 Hz and 20 Hz.

[0132] In an embodiment, a computer program product, storable in a control unit (56) and designed in such a way that, when executed in the control unit (56), the control unit (56) becomes configured to perform a measuring method (80).

[0133] In an embodiment, a device comprises: an acceleration sensor, which, in operation, generates an acceleration signal indicative of vibrations associated with a cartridge of a laser printer; a charge variation sensor, which, in operation, generates a charge-variation signal indicative of a variation of electrostatic charge associated with the cartridge of the laser printer; and processing circuitry coupled to the acceleration sensor and to the charge variation sensor; wherein the processing circuitry, in operation: generates feature parameters based on the acceleration signal and the charge-variation signal; and generates a level signal indicative of a level of toner in the cartridge based on the generated feature parameters.

[0134] In an embodiment, the processing circuitry comprises: a feature extraction module, which, in operation, generates the feature parameters; and a classification module coupled to the feature extraction module, wherein, in operation, the classification module generates the level signal.

[0135] In an embodiment, the device comprises an integrated circuit, the integrated circuit including: the processing circuitry; the processing circuitry and the acceleration sensor; the processing circuitry and the charge variation sensor; or the processing circuitry, the acceleration sensor and the charge variation sensor.

[0136] In an embodiment, the processing circuitry, in operation, determines a norm of the acceleration signal, generating a normalized acceleration signal, and generates the feature parameters based on the normalized acceleration signal.

[0137] In an embodiment, the charge variation sensor comprises a filter, which, in operation, filters the charge-variation signal, generating a filtered charge-variation signal, and the processing circuitry, in operation, generates the feature parameters based on the filtered charge-variation signal. In an embodiment, the filter is a band-pass filter.

[0138] In an embodiment, the acceleration sensor comprises a filter, which, in operation, filters the acceleration signal, generating a filtered acceleration signal, and the processing circuitry, in operation, generates the feature parameters based on the filtered acceleration signal.

[0139] In an embodiment, a system comprises: a laser printer cartridge having an external body; and toner-level measurement circuitry coupled to the external body, wherein the toner-level measurement circuitry, in operation: generates an acceleration signal indicative of vibrations associated with the laser printer cartridge; generates a charge-variation signal indicative of a variation of electrostatic charge associated with the laser printer cartridge; generates feature parameters based on the acceleration signal and the charge-variation signal; and generates a level signal indicative of a level of toner in the laser printer cartridge based on the generated feature parameters.

[0140] In an embodiment, the toner-level measurement circuitry comprises: an acceleration sensor, which, in operation, generates the acceleration signal; a charge variation sensor, which, in operation, generates the charge-variation signal; and processing circuitry coupled to the acceleration sensor and to the charge variation sensor, wherein the processing circuitry, in operation: generates the feature parameters based on the acceleration signal and the charge-variation signal; and generates the level signal indicative of a level of toner in the cartridge based on the generated feature parameters.

[0141] In an embodiment, the laser printer cartridge includes: a toner hopper housed in the external body; and rotating elements housed in the external body, the rotating elements including: a developer roller; an OPC drum; a primary charge roller; or combinations thereof.

[0142] In an embodiment, the acceleration sensor is coupled to the external body adjacent to a rotating element of the rotating elements; and the charge variation sensor is coupled to the external body adjacent to the toner hopper.

[0143] In an embodiment, the system comprises a laser printer, the laser printer including the laser printer cartridge and the toner-level measurement circuitry.

[0144] In an embodiment, a method comprises: generating an acceleration signal indicative of vibrations associated with a laser printer cartridge; generating a charge-variation signal indicative of a variation of electrostatic charge associated with the laser printer cartridge; generating feature parameters based on the acceleration signal and the charge-variation signal; and generating a level signal indicative of a level of toner in the laser printer cartridge based on the generated feature parameters.

[0145] In an embodiment, the method comprises: generating the acceleration sensor using an acceleration sensor of the laser printer cartridge; generating the charge-variation signal using a charge variation sensor of the laser printer cartridge; generating the feature parameters using processing circuitry of the laser printer cartridge, the processing circuitry being coupled to the acceleration sensor and to the charge variation sensor; and generating the level signal using the processing circuitry.

[0146] In an embodiment, the generating the level signal comprises selecting a value from a set of values corresponding to defined levels of toner in the laser printer cartridge.

[0147] In an embodiment, the generating the level signal comprises processing the feature parameters using a classifier. In an embodiment, the classifier is a decision-tree classifier.

[0148] In an embodiment, the generating the feature parameters comprises determining: a mean of the acceleration signal, a mean of the charge-variation signal, a variance of the acceleration signal, a variance of the charge-variation signal, an energy of the acceleration signal, an energy of the charge-variation signal, a zero cross value of the acceleration signal, a zero cross value of the

charge-variation signal, a peak detector value of the acceleration signal, a peak detector value of the charge-variation signal, a peak-to-peak distance of the acceleration signal, a peak-to-peak distance of the charge-variation signal, a minimum value of the acceleration signal, a minimum value of the charge-variation signal, a maximum value of the acceleration signal, a maximum value of the charge-variation signal, or combinations thereof.

[0149] In an embodiment, the method comprises: generating a normalized acceleration signal; and generating feature parameters based on the normalized acceleration signal.

[0150] In an embodiment, the method comprises: generating a filtered charge-variation signal; and generating feature parameters based on the filtered charge-variation signal.

[0151] In an embodiment, generating the filtered charge-variation signal comprises band-pass filtering the charge-variation signal.

[0152] In an embodiment, a non-transitory computer-readable medium's contents configure processing circuitry to perform a method, the method comprising: generating an acceleration signal indicative of vibrations associated with a laser printer cartridge; generating a charge-variation signal indicative of a variation of electrostatic charge associated with the laser printer cartridge; generating feature parameters based on the acceleration signal and the charge-variation signal; and generating a level signal indicative of a level of toner in the laser printer cartridge based on the generated feature parameters. In an embodiment, the contents comprise instructions executable by the processing circuitry. In an embodiment, the processing circuitry comprises a classifier; and the contents configure the classifier to generate level signal based on the feature parameters.

[0153] Some embodiments may take the form of or comprise computer program products. For example, according to one embodiment there is provided a computer readable medium comprising a computer program adapted to perform one or more of the methods or functions described above. The medium may be a physical storage medium, such as for example a Read Only Memory (ROM) chip, or a disk such as a Digital Versatile Disk (DVD-ROM), Compact Disk (CD-ROM), a hard disk, a memory, a network, or a portable media article to be read by an appropriate drive or via an appropriate connection, including as encoded in one or more barcodes or other related codes stored on one or more such computer-readable mediums and being readable by an appropriate reader device.

[0154] Furthermore, in some embodiments, some or all of the methods and/or functionality may be implemented or provided in other manners, such as at least partially in firmware and/or hardware, including, but not limited to, one or more application-specific integrated circuits (ASICs), digital signal processors, discrete circuitry, logic gates, standard integrated circuits, controllers (e.g., by executing appropriate instructions, and including microcontrollers and/or embedded controllers), field-programmable gate arrays (FPGAs), complex programmable logic devices (CPLDs), etc., as well as devices that employ RFID technology, and various combinations thereof.

[0155] The various embodiments described above can be combined to provide further embodiments. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.

[0156] These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

## Claims

1. A device, comprising: an acceleration sensor, which, in operation, generates an acceleration signal indicative of vibrations associated with a cartridge of a laser printer; a charge variation sensor, which, in operation, generates a charge-variation signal indicative of a variation of

electrostatic charge associated with the cartridge of the laser printer; and processing circuitry coupled to the acceleration sensor and to the charge variation sensor; wherein the processing circuitry, in operation: generates feature parameters based on the acceleration signal and the charge-variation signal; and generates a level signal indicative of a level of toner in the cartridge based on the generated feature parameters, the generated feature parameters including a peak-to-peak distance of the acceleration signal and a maximum of the charge variation signal.

**2.** The device according to claim 1, wherein the processing circuitry comprises: a feature extraction module, which, in operation, generates the feature parameters; and a classification module coupled to the feature extraction module, wherein, in operation, the classification module generates the level signal.

**3.** The device according to claim 1, comprising an integrated circuit, the integrated circuit including: the processing circuitry; the processing circuitry and the acceleration sensor; the processing circuitry and the charge variation sensor; or the processing circuitry, the acceleration sensor and the charge variation sensor.

**4.** The device according to claim 1, wherein, the processing circuitry, in operation, determines a norm of the acceleration signal, generating a normalized acceleration signal, and generates the feature parameters based on the normalized acceleration signal.

**5.** The device according to claim 1, wherein, the charge variation sensor comprises a filter, which, in operation, filters the charge-variation signal, generating a filtered charge-variation signal, and the processing circuitry, in operation, generates the feature parameters based on the filtered charge-variation signal.

**6.** The device of claim 5, wherein the filter is a band-pass filter.

**7.** The device according to claim 1, wherein, the acceleration sensor comprises a filter, which, in operation, filters the acceleration signal, generating a filtered acceleration signal, and the processing circuitry, in operation, generates the feature parameters based on the filtered acceleration signal.

**8.** A system, comprising: a laser printer cartridge having an external body; and toner-level measurement circuitry coupled to the external body, wherein the toner-level measurement circuitry, in operation: generates an acceleration signal indicative of vibrations associated with the laser printer cartridge; generates a charge-variation signal indicative of a variation of electrostatic charge associated with the laser printer cartridge; generates feature parameters based on the acceleration signal and the charge-variation signal; and generates a level signal indicative of a level of toner in the laser printer cartridge based on the generated feature parameters, the generated feature parameters including a peak-to-peak distance of the acceleration signal and a maximum of the charge variation signal.

**9.** The system according to claim 8, wherein the toner-level measurement circuitry comprises: an acceleration sensor, which, in operation, generates the acceleration signal; a charge variation sensor, which, in operation, generates the charge-variation signal; and processing circuitry coupled to the acceleration sensor and to the charge variation sensor, wherein the processing circuitry, in operation: generates the feature parameters based on the acceleration signal and the charge-variation signal; and generates the level signal indicative of a level of toner in the cartridge based on the generated feature parameters.

**10.** The system according to claim 9, wherein the laser printer cartridge includes: a toner hopper housed in the external body; and rotating elements housed in the external body, the rotating elements including: a developer roller; an OPC drum; a primary charge roller; or combinations thereof.

**11.** The system according to claim 10, wherein, the acceleration sensor is coupled to the external body adjacent to a rotating element of the rotating elements; and the charge variation sensor is coupled to the external body adjacent to the toner hopper.

**12.** The system according to claim 8, comprising a laser printer, the laser printer including the laser



printer cartridge and the toner-level measurement circuitry.

**13.** A method, comprising: generating an acceleration signal indicative of vibrations associated with a laser printer cartridge; generating a charge-variation signal indicative of a variation of electrostatic charge associated with the laser printer cartridge; generating feature parameters based on the acceleration signal and the charge-variation signal; and generating a level signal indicative of a level of toner in the laser printer cartridge based on the generated feature parameters, the generated feature parameters including a peak-to-peak distance of the acceleration signal and a maximum of the charge variation signal.

**14.** The method according to claim 13, comprising: generating the acceleration sensor using an acceleration sensor of the laser printer cartridge; generating the charge-variation signal using a charge variation sensor of the laser printer cartridge; generating the feature parameters using processing circuitry of the laser printer cartridge, the processing circuitry being coupled to the acceleration sensor and to the charge variation sensor; and generating the level signal using the processing circuitry.

**15.** The method according to claim 13, wherein the generating the level signal comprises selecting, a value from a set of values corresponding to defined levels of toner in the laser printer cartridge.

**16.** The method according to claim 15, wherein the generating the level signal comprises processing the feature parameters using a classifier.

**17.** The method according to claim 16, wherein the classifier is a decision-tree classifier.

**18.** The method according to claim 13, wherein generating the feature parameters comprises determining: a mean of the acceleration signal, a mean of the charge-variation signal, a variance of the acceleration signal, a variance of the charge-variation signal, an energy of the acceleration signal, an energy of the charge-variation signal, a zero cross value of the acceleration signal, a zero cross value of the charge-variation signal, a peak detector value of the acceleration signal, a peak detector value of the charge-variation signal, a peak-to-peak distance of the charge-variation signal, a minimum value of the acceleration signal, a minimum value of the charge-variation signal, a maximum value of the acceleration signal, or combinations thereof.

**19.** The method according to claim 13, comprising: generating a normalized acceleration signal; and generating feature parameters based on the normalized acceleration signal.

**20.** The method according to 13, comprising: generating a filtered charge-variation signal; and generating feature parameters based on the filtered charge-variation signal.

**21.** The method according to claim 20, wherein generating the filtered charge-variation signal comprises band-pass filtering the charge-variation signal.

**22.** The method according to claim 17, wherein, the plurality of defined levels of toner in the cartridge comprises a first to a fifth level, the first level is greater than the second level, that is greater than the third level, that is greater than the fourth level, that is greater than the fifth level, the generating the level signal comprises: checking whether the peak-to-peak distance satisfies a first condition, the first condition being satisfied if the peak-to-peak distance is lower than, or equal to, a first threshold distance; if the first condition is satisfied, checking if the peak-to-peak distance satisfies a second condition, the second condition being satisfied if the peak-to-peak distance is lower than, or equal to, a second threshold distance; if the second condition is satisfied, determining the level signal as corresponding to the first level; if the second condition is not satisfied, checking if the maximum satisfies a third condition, the third condition being satisfied if the maximum is lower than, or equal to, a first threshold maximum; if the third condition is not satisfied, determining the level signal as corresponding to the third level; if the third condition is satisfied, checking if the peak-to-peak distance satisfies a fourth condition, the fourth condition being satisfied if the peak-to-peak distance is lower than, or equal to, a third threshold distance; if the fourth condition is satisfied, determining the level signal as corresponding to the third level; if the fourth condition is not satisfied, determining the level signal as corresponding to the second level; if the first condition is not satisfied, checking if the peak-to-peak distance satisfies a fifth

condition, the fifth condition being satisfied if the peak-to-peak distance is lower than, or equal to, a fourth threshold distance; if the fifth condition is satisfied, determining the level signal as corresponding to the fourth level; and if the fifth condition is not satisfied, determining the level signal as corresponding to the fifth level, and the second threshold distance (A2) is lower than the third threshold distance (A3), that is lower than the first threshold distance (A1), that is lower than the fourth threshold distance (A4).

**23.** The method according to claim 22, wherein: the first threshold distance (A1) is equal to  $0.306 \pm 0.001$  mg; the second threshold distance (A2) is equal to  $0.141 \pm 0.001$  mg; the first threshold maximum (C1) is equal to  $1.463 \pm 0.001$  LSB; the third threshold distance (A3) is equal to  $0.193 \pm 0.001$  mg; and the fourth threshold distance (A4) is equal to  $0.460 \pm 0.001$  mg.

**24.** A non-transitory computer-readable medium having contents which configure processing circuitry to perform a method, the method comprising: generating an acceleration signal indicative of vibrations associated with a laser printer cartridge; generating a charge-variation signal indicative of a variation of electrostatic charge associated with the laser printer cartridge; generating feature parameters based on the acceleration signal and the charge-variation signal; and generating a level signal indicative of a level of toner in the laser printer cartridge based on the generated feature parameters, the generated feature parameters including a peak-to-peak distance of the acceleration signal and a maximum of the charge variation signal.

**25.** The non-transitory computer-readable medium of claim 24, wherein the contents comprise instructions executable by the processing circuitry.

**26.** The non-transitory computer-readable medium of claim 24, wherein, the processing circuitry comprises a classifier; and the contents configure the classifier to generate level signal based on the feature parameters.

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